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https://github.com/julianmak/OCES4303_ML_ocean

The repository principally contains the compiled products rather than the source for size reasons.

- ▶ Associated Python code (as Jupyter notebooks mostly) will be held on the same repository. The source data however might be big, so I am going to be naughty and possibly just refer you to where you might get the data if that is the case (e.g. JRA-55 data). I know I should make properly reproducible binders etc., but I didn't...
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OCES 4303 :
an introduction to **data-driven and ML methods** in ocean sciences

Session 6: decision trees

Outline

- ▶ anatomy of a decision tree
- ▶ some concepts of probability
→ information entropy and information gain
- ▶ usage of decision trees as classifiers or regressors
- ▶ demonstration: penguins data



Figure: An example of a tree (a broc-collie).

Oceanic application

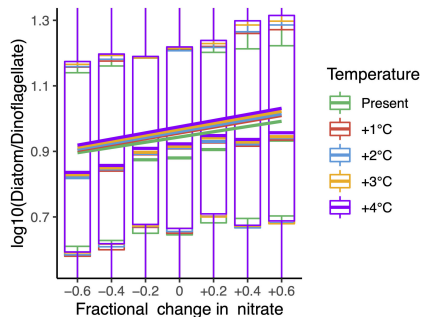


Figure: Predicted log of diatom-dinoflagellate ratio in HK coastal waters using random forests (which is a collection of decision trees) under projected warming scenarios. From Fig. 4 of Cheung *et al.* (2021).

- marine pollution
 - used station data to train up a regression model
 - focused on diatom-dinoflagellate ratio
 - identified key features of importance
 - can do projection (!!!) and find pollution increases under warming

(see also Lee *et al.* (2025) from OCES)

Anatomy

- ▶ **root:** head node
- ▶ **leaves:** terminal nodes
- ▶ **nodes:** the boxes
- ▶ **branches:** connectors of nodes with decision
 - strictly speaking the 'yes' and 'no' should be branches
- ▶ **levels/depth:** how far you are from root
 - here it could be tree of depth 1 or 2
- ▶ **parent/child:** to talk about nodes, parent is one level closer to root



My Cat's Decision-Making Tree.

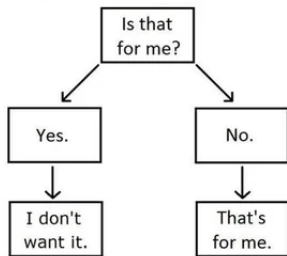


Figure: Example of a decision tree.

Recap: probability

- ▶ main goal: how do we decide on the criterion to split data (i.e. branching)?
→ want the data to tell us, rather than us doing it manually

Recap: probability

- ▶ main goal: how do we decide on the criterion to split data (i.e. branching)?
→ want the data to tell us, rather than us doing it manually
- ▶ need some concepts in **probability**, do this through examples
→ e.g. for a **fair** 6-side dice, the possible **events** are

$$X = \{1, 2, 3, 4, 5, 6\},$$

and assigned to these events are **probabilities** $p_i \in [0, 1]$

→ all probabilities should sum to 1, i.e. $\sum_i p_i = 1$

→ **fair** $\Rightarrow$ uniform distribution $\Rightarrow p_i = 1/N = 1/6$

Recap: probability

- ▶ there is a measure called the **information entropy** defined as

$$H = \sum_i H_i = - \sum_i p_i \log_a p_i$$

- called **Shannon entropy/index** (from ecology)
- base a is usually 2, e or 10; the value itself doesn't actually matter too much (I will take $a = e$ so $\log = \ln$)

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- ▶ measure of species diversity (in ecology); think of it as measure of **surprise**, e.g.
 - if $X = \{1, 1, 1, 1, 1, 1\}$, then there is no 'surprise' on what you should draw ($H = 0$)
 - if $X = \{1, 2, 3, 4, 5, 6\}$, events are maximally different and maximally 'surprising', ($H = \log N = \log 6$ here)

Recap: probability

- ▶ from that, the **information gain** is defined as

$$\text{IG} = H_p - \sum_{i=0}^N p_{c,i} H_{c,i}$$

→ think entropy of parent class minus the weighted averages of entropy of child class

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- ▶ e.g., if I have cats and dogs of various colours, hair length...
 - my parent class could be 'cats' + 'dogs' (2 classes)
 - if I segment on **three** 'colours', then I want
 - ▶ probabilities of the resulting **six** classes (probably)
 - ▶ entropy of that

Recap: probability

$$\text{IG} = H_p - \sum_{i=0}^N p_{c,i} H_{c,i}$$

- ▶ positive values mean entropy of data at the next level has **decreased**
 - data is getting more 'pure'
 - if proceeding to $H = 0$ then we have maximum purity with no uncertainty
 - no uncertainty = max information, and information **gain**
- ▶ want to split according to maximum information gain
(or maximise entropy decrease)

Recap: probability

- ▶ **Gini index** measure a similar thing to entropy, and is defined as

$$G = \sum_i G_i = \sum_i p_i(1 - p_i)$$

→ single species gives $G = 0$, maximally pure

- ▶ can do a similar thing to information gain to minimise the Gini index with splitting
 - used more in decision trees (computationally faster)
 - entropy measures used more in things like neural networks

Demonstration: penguins data

- consider **classification** problem first
→ predict species from other features

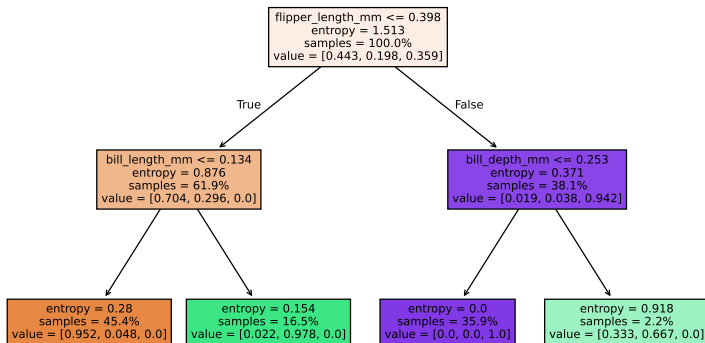


Figure: Tree classifier for penguins data. Maximum depth 2 and uses information entropy criterion.

Demonstration: penguins data

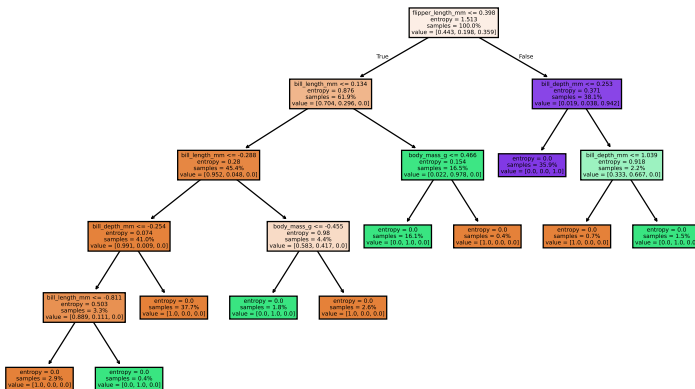


Figure: Tree classifier for penguins data. No maximum depth, and uses information entropy criterion.

Demonstration: penguins data

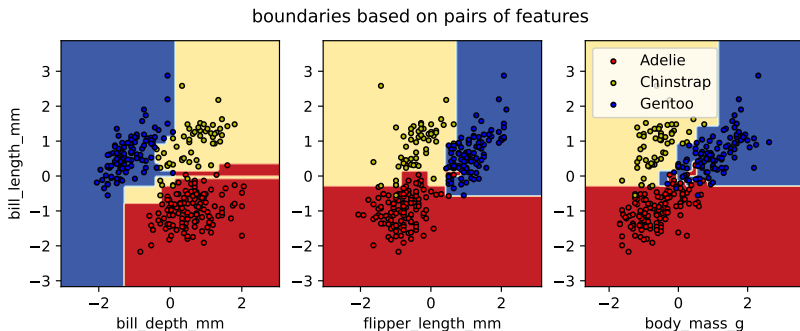


Figure: Tree classifier decision boundaries based on two features for penguins data.

- notice decision trees basically do piecewise constant predictions (see this again when doing regression)

Demonstration: penguins data

- consider now the **regression** problem
 - same problem identified in the second session
 - include species as an input feature?

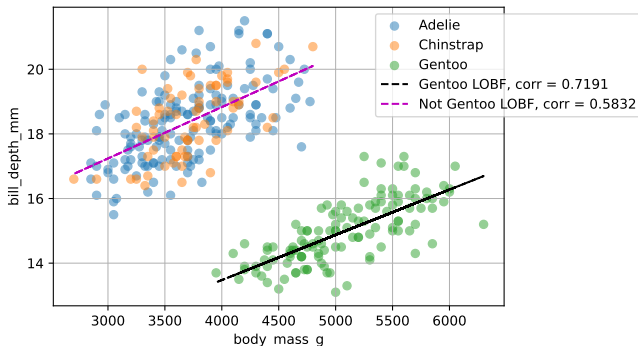


Figure: Instead of models for each species, one model that does different decisions based on species?

Demonstration: penguins data

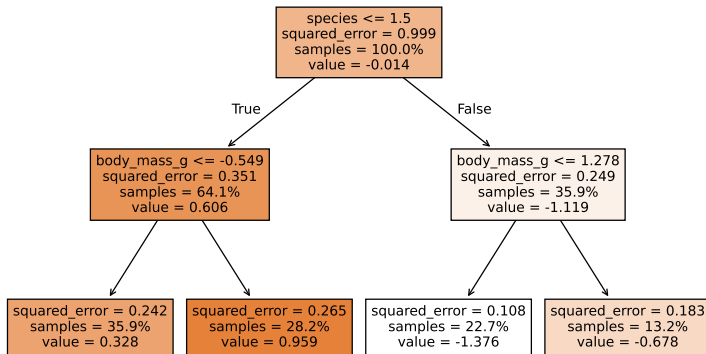


Figure: Tree regressor for penguins data. Maximum depth two, squared loss.

Demonstration: penguins data

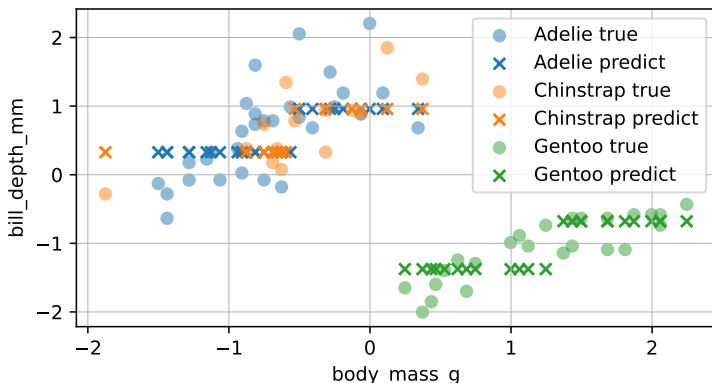


Figure: Tree regressor for penguins data. Maximum depth two, squared loss.

- notice the species predictions are clustered accordingly
→ notice the piecewise constant predictions (model too shallow)

Demonstration: penguins data

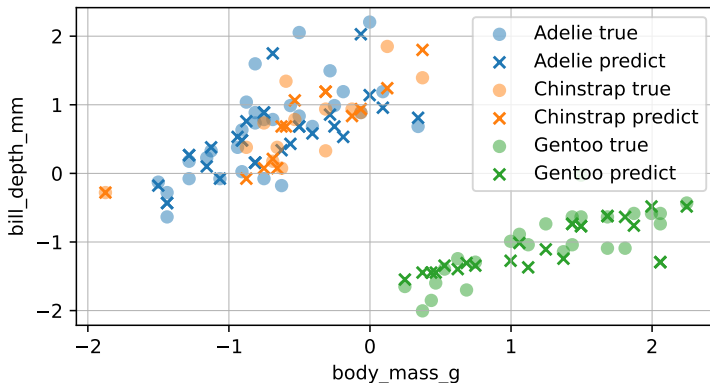


Figure: Tree regressor for penguins data. No maximum depth, squared loss.

- recover variability by having more complexity

Demonstration: penguins data

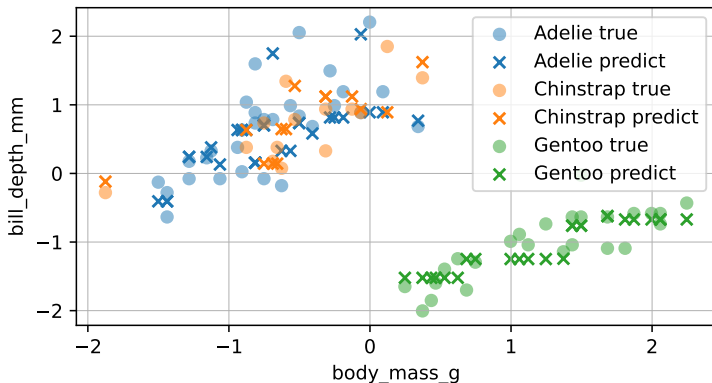


Figure: Tree regressor for penguins data with post-pruning. No maximum depth, squared loss.

- **pruning** to avoid over-fitting + promote robustness
→ remove nodes that add extra complexity but not that much skill (cf. A/BIC)

Demonstration

- ▶ went into some detail in how decision trees are formed
 - concepts in information entropy
 - used also with model selection and neural networks
- ▶ trees as classifiers and/or regressors
 - need for cross-validation / hyper-parameter tuning
 - ensembles? (see next lec)

Moving to a Jupyter notebook →



Figure: An example of a tree (a broc-collie).